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1. Write C programs to implement following disk scheduling algorithms:

(a) FCFS

(b) SCAN

(c) SSTF

Run your program to consider the following scenario:

Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at

cylinder 95, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105,

110, 135 and 145. Starting from the current head position, Find the the total distance (in

cylinders) that the disk arm moves to satisfy all the pending requests for each of the above

disk-scheduling algorithms.

Assignment - 10

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Q1)

(a) FCFS

```
#include <stdio.h>
#include <stdlib.h>
```

```
int main() {
```

```
    int requests[] = {30, 85, 90, 100, 105, 110, 135, 145};
```

```
    int n = sizeof(requests) / sizeof(requests[0]);
```

```
    int head = 95;
```

```
    int total_movement = 0;
```

```
    printf("FCFS Disk Scheduling\n");
```

```
    printf("Initial head position: %d\n", head);
```

```
    printf("Seek Sequence: %d", head);
```

```
    for (int i = 0; i < n; i++) {
```

```
        total_movement += abs(head - requests[i]);
```

```
        head = requests[i];
```

```
    }
    printf("→ %d", head);
```

```
    printf("\n Total head movement = %d\n",
```

```
        total_movement);
```

```
    return 0;
```

```
}
```


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(b) SCAN

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int compare (const void *a, const void *b) {  
    return (*(int*)a - *(int*)b);  
}
```

```
}
```

```
int main() {
```

```
    int requests[] = {30, 85, 90, 100, 105, 110, 135,  
                      145};
```

```
    int n = sizeof(requests) / sizeof(requests[0]);
```

```
    int head = 95;
```

```
    int disk-size = 200;
```

```
    int total-movement = 0;
```

```
    int temp[n+1];
```

```
    for (int i=0; i<n; i++) {  
        temp[i] = requests[i];  
    }
```

```
}
```

```
temp[n] = head;
```

```
qsat(temp, n+1, sizeof(int), compare);
```

```
int head-index = 0;
```



```

for (int i=0; i<n; i++) {
    if (temp[i] == head) {
        head_index = i;
        break;
    }
}

```

```

printf("SCAN Disk scheduling\n");
printf("Initial head position : %d\n", head);
printf("Seek Sequence : %d", head);

```

```

int current = head;

```

```

for (int i = head_index + 1; i <= n; i++) {
    total_movement += abs(current - temp[i]);
    current = temp[i];
    printf("→ %d", current);
}

```

```

if (current != disk_size) {
    total_movement += abs(current - disk_sizetemp[i]);
    current = temp[i] + disk_size;
    printf("→ %d", current);
}

```

```

for (int i = head_index - 1; i >= 0; i--) {
    total_movement += abs(current - temp[i]);
    current = temp[i];
    printf("→ %d", current);
}

```



```
printf("\n Total head movement = %d\n",  
total-movement);
```

```
return 0;
```

```
}
```

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(C) SSTF

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <limits.h>
```

```
int main () {
```

```
int requests[] = {30, 85, 90, 100, 105, 110, 135,  
145};
```

```
int n = sizeof(requests) / sizeof(requests[0]);
```

```
int done[n] = {0};
```

```
int head = 95;
```

```
int total-movement = 0;
```

```
printf(" SSTF Disk scheduling\n");
```

```
printf(" Initial head position: %d\n", head);
```

```
printf(" Seek Sequence = %d", head);
```


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```
for(int count=0; count<n; count++) {
    int min-distance = INT_MAX;
    int index = -1;
    for(int i=0; i<n; i++) {
        if(!done[i]) {
            int distance = abs(head-requests[i]);
            if(distance < min-distance) {
                min-distance = distance;
                index = i;
            }
        }
    }
    done[index] = 1;
    total-movement += min-distance;
    head = requests[index];
    printf("→ %d", head);
}
printf("\n Total head movement = %d\n",
        total-movement);

return 0;
}
```

Output:

```
user@hp-HP-EliteDesk-800-G1-SFF:~/deep.20233517$ gcc q1a.c
user@hp-HP-EliteDesk-800-G1-SFF:~/deep.20233517$ ./a.out
FCFS Disk Scheduling
Initial head position: 95
Seek sequence: 95 -> 30 -> 85 -> 90 -> 100 -> 105 -> 110 -> 135 -> 145
Total head movement = 180
user@hp-HP-EliteDesk-800-G1-SFF:~/deep.20233517$ gcc q1b.c
user@hp-HP-EliteDesk-800-G1-SFF:~/deep.20233517$ ./a.out
SCAN Disk Scheduling
Initial head position: 95
Seek sequence: 95 -> 100 -> 105 -> 110 -> 135 -> 145 -> 200 -> 90 -> 85
Total head movement = 275
user@hp-HP-EliteDesk-800-G1-SFF:~/deep.20233517$ gcc q1c.c
user@hp-HP-EliteDesk-800-G1-SFF:~/deep.20233517$ ./a.out
SSTF Disk Scheduling
Initial head position: 95
Seek sequence: 95 -> 90 -> 85 -> 100 -> 105 -> 110 -> 135 -> 145 -> 30
Total head movement = 185
```